

# Hermitian line bundles

Let  $K$  be a number field with ring of integers  $\mathcal{O}_K$ . Let  $\mathcal{X}$  be an arithmetic variety over  $\mathrm{Spec} \mathcal{O}_K$  of relative dimension  $n$ . **Yang:** flat, separated, of finite type,

For every  $\sigma : \mathcal{O}_K \rightarrow \mathbb{C}$ , we denote by  $\mathcal{X}_\sigma(\mathbb{C}) := \mathcal{X} \times_{\mathrm{Spec} \mathcal{O}_K, \sigma} \mathrm{Spec} \mathbb{C}$  the complex analytic space associated to the base change of  $\mathcal{X}$  along  $\sigma$ .

## 1 Definitions and examples

**Definition 1.** A hermitian line bundle  $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$  on  $\mathcal{X}$  consists of

- a line bundle  $\mathcal{L}$  on  $\mathcal{X}$ ;
- for each embedding  $\sigma : K \hookrightarrow \mathbb{C}$  (equivalently, each infinite place  $\sigma \in M_K^\infty$ ), a smooth hermitian metric  $\|\cdot\|_\sigma$  on the complex line bundle  $\mathcal{L}_\sigma(\mathbb{C})$  over  $\mathcal{X}_\sigma(\mathbb{C})$ , which is invariant under the complex conjugation map.

**Yang:** To be checked.

**Example 2.** Consider  $\mathcal{X} = \mathbb{P}_{\mathcal{O}_K}^n$  and  $\mathcal{L} = \mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)$ . There is a class of natural hermitian metrics on  $\mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)_{\sigma}(\mathbb{C})$  called the *Fubini-Study metrics* constructed as follows.

Recall that  $\mathbb{P}^n$  represents the functor that sends a scheme  $S$  to the set of isomorphism classes of pairs  $(\mathcal{E}, \phi)$ , where  $\mathcal{E}$  is a rank  $n+1$  vector bundle on  $S$  and  $\phi : \mathcal{E} \rightarrow \mathcal{L}$  is a surjective morphism onto a line bundle  $\mathcal{L}$  on  $S$ . And the universal pair on  $\mathbb{P}^n$  is given by  $\mathcal{O}_{\mathbb{P}^n}^{\oplus n+1} \rightarrow \mathcal{O}_{\mathbb{P}^n}(1)$ .

For each  $\sigma \in M_K^\infty$ , fix a hermitian metric  $\|\cdot\|$  on  $\mathbb{C}^{n+1}$ . Every point  $x \in \mathbb{P}_{\mathcal{O}_K}^n(\sigma)$ , it corresponds to a surjective morphism  $\phi_x : \mathbb{C}^{n+1} \rightarrow L_x$  with  $L_x$  the fiber of  $\mathcal{L}_\sigma(\mathbb{C})$  at  $x$ . Explicitly, every point  $x = [z_0 : z_1 : \cdots : z_n] \in \mathbb{P}^n(\mathbb{C})$  corresponds to the surjective morphism

$$\phi_{[z_0:z_1:\dots:z_n]} : \mathbb{C}^{n+1} \rightarrow \mathbb{C} \cong L_x, \quad (x_0, x_1, \dots, x_n) \mapsto x_0 z_0 + x_1 z_1 + \cdots + x_n z_n.$$

The fixed hermitian metric induces the quotient hermitian metric  $\|\cdot\|_x$  on  $L_x$ . Hence we obtain a hermitian metric  $\|\cdot\|_\sigma$  on  $\mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)_{\sigma}(\mathbb{C})$ . This construction defines a hermitian line bundle  $\overline{\mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)}$ .

**Yang:** To be checked.

**Definition 3.** Let  $\overline{\mathcal{L}}_1$  and  $\overline{\mathcal{L}}_2$  be hermitian line bundles on  $\mathcal{X}$ .

- The *tensor product*  $\overline{\mathcal{L}}_1 \otimes \overline{\mathcal{L}}_2$  is defined as the line bundle  $\mathcal{L}_1 \otimes \mathcal{L}_2$  on  $\mathcal{X}$  equipped with the metrics  $\|s_1 \otimes s_2\|_\sigma = \|s_1\|_{\sigma,1} \cdot \|s_2\|_{\sigma,2}$  on  $(\mathcal{L}_1 \otimes \mathcal{L}_2)_\sigma(\mathbb{C})$
- The *dual*  $\overline{\mathcal{L}}_1^\vee$  is defined as the dual line bundle  $\mathcal{L}_1^\vee$  on  $\mathcal{X}$  equipped with the dual metrics  $\|\cdot\|_{\sigma,1}^\vee$  on  $(\mathcal{L}_1^\vee)_\sigma(\mathbb{C})$  for each  $\sigma \in M_K^\infty$ .

**Yang:** To be checked.

**Proposition 4.** The set of isomorphism classes of hermitian line bundles on  $\mathcal{X}$  forms an abelian group under the tensor product operation. The group is called the *arithmetic Picard group* of  $\mathcal{X}$  and is denoted by  $\widehat{\text{Pic}}(\mathcal{X})$ .

*Proof.* Direct verification.  $\square$

Note that for every  $\sigma \in M_K^\infty$ , any two smooth hermitian metrics on  $\mathcal{L}_\sigma(\mathbb{C})$  differ by a smooth function on  $\mathcal{X}_\sigma(\mathbb{C})$ . Explicitly, for any two smooth hermitian metrics  $\|\cdot\|_\sigma$  and  $\|\cdot\|'_\sigma$  on  $\mathcal{L}_\sigma(\mathbb{C})$ , there exists a unique smooth function  $f : \mathcal{X}_\sigma(\mathbb{C}) \rightarrow \mathbb{R}$  such that

$$\|s\|'_\sigma = e^{-f} \|s\|_\sigma$$

for all local sections  $s$  of  $\mathcal{L}_\sigma(\mathbb{C})$ . Conversely, given any smooth hermitian metric  $\|\cdot\|_\sigma$  on  $\mathcal{L}_\sigma(\mathbb{C})$  and any smooth function  $f : \mathcal{X}_\sigma(\mathbb{C}) \rightarrow \mathbb{R}$ , we can define a new smooth hermitian metric  $\|\cdot\|'_\sigma$  on  $\mathcal{L}_\sigma(\mathbb{C})$  by the above formula.

**Example 5.** Let  $\mathcal{X} = \mathbb{P}_\mathbb{Z}^n$ . Then we have

$$\widehat{\text{Pic}}(\mathbb{P}_\mathbb{Z}^n) \cong \mathbb{Z} \times \mathcal{C}^\infty(\mathbb{P}^n(\mathbb{C}), \mathbb{R}).$$

Here the generator of the first factor  $\mathbb{Z}$  corresponds to the hermitian line bundle  $\overline{\mathcal{O}_{\mathbb{P}_\mathbb{Z}^n}(1)}$  constructed in [Example 2](#).

**Definition 6.** Let  $f : \mathcal{Y} \rightarrow \mathcal{X}$  be a morphism of arithmetic varieties over  $\text{Spec } \mathcal{O}_K$ . Let  $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$  be a hermitian line bundle on  $\mathcal{X}$ . The *pullback*  $f^*\overline{\mathcal{L}}$  is defined as the line bundle  $f^*\mathcal{L}$  on  $\mathcal{Y}$  equipped with the metrics  $\|s\|'_\sigma = \|f_\sigma^* s\|_\sigma$  on  $(f^*\mathcal{L})_\sigma(\mathbb{C})$  for each  $\sigma \in M_K^\infty$ , where  $f_\sigma : \mathcal{Y}_\sigma(\mathbb{C}) \rightarrow \mathcal{X}_\sigma(\mathbb{C})$  is the induced morphism of complex manifolds. **Yang: To be checked.**

**Definition 7.** Let  $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$  be a hermitian line bundle on  $\mathcal{X}$ . The set of *small sections* of  $\overline{\mathcal{L}}$  is defined as

$$H^0(\mathcal{X}, \overline{\mathcal{L}}) := \{s \in H^0(\mathcal{X}, \mathcal{L}) : \|s(x)\|_\sigma \leq 1, \forall \sigma \in M_K^\infty, x \in \mathcal{X}_\sigma(\mathbb{C})\}.$$

**Yang: To be checked.**

**Example 8.** Let  $\mathcal{X} = \mathbb{P}_\mathbb{Z}^n$  and  $\overline{\mathcal{L}} = \overline{\mathcal{O}_{\mathbb{P}_\mathbb{Z}^n}(1)}$  be the hermitian line bundle constructed in [Example 2](#). We have

$$H^0(\mathbb{P}_\mathbb{Z}^n, \mathcal{L}) = \mathbb{Z}X_0 \oplus \mathbb{Z}X_1 \oplus \cdots \oplus \mathbb{Z}X_n,$$

where  $X_0, X_1, \dots, X_n$  are the standard homogeneous coordinates on  $\mathbb{P}_\mathbb{Z}^n$ . For every  $s = \sum a_n X_n \in H^0(\mathbb{P}_\mathbb{Z}^n, \mathcal{L})$  and  $x = [x_0 : \cdots : x_n] \in \mathbb{P}^n(\mathbb{C})$ , we have

$$\|s(x)\| = \inf_{x_0 y_0 + \cdots + x_n y_n = 0} \left( \sum |a_k + y_k|^2 \right)^{1/2} = \frac{|\sum a_k x_k|}{\|x\|},$$

where  $\|x\| = (|x_0|^2 + |x_1|^2 + \cdots + |x_n|^2)^{1/2}$ . When  $x$  varies, by Cauchy-Schwarz inequality, we have

$$\sup_{x \in \mathbb{P}^n(\mathbb{C})} \|s(x)\| = \sup_{x \in \mathbb{P}^n(\mathbb{C})} \frac{|\sum a_k x_k|}{\|x\|} = \|a\|.$$

Note that  $a_n \in \mathbb{Z}$  for all  $n$ . Therefore, we have

$$H^0(\mathbb{P}_\mathbb{Z}^n, \overline{\mathcal{O}_{\mathbb{P}_\mathbb{Z}^n}(1)}) = \{0, \pm X_0, \pm X_1, \dots, \pm X_n\}$$

and

$$\#H^0(\mathbb{P}_{\mathbb{Z}}^n, \overline{\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(1)}) = 2n + 3.$$

**Proposition 9.** The set  $H^0(\mathcal{X}, \overline{\mathcal{L}})$  is finite.

**Definition 10.** Let  $\overline{\mathcal{L}}$  be a hermitian line bundle on  $\mathcal{X}$ . We set

$$h^0(\mathcal{X}, \overline{\mathcal{L}}) := \log \#H^0(\mathcal{X}, \overline{\mathcal{L}}).$$

The *volume* of  $\overline{\mathcal{L}}$  is defined as

$$\text{vol}(\overline{\mathcal{L}}) := \limsup_{m \rightarrow \infty} \frac{h^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes m})}{m^{n+1}/(n+1)!}.$$

**Example 11.** We continue with [Example 8](#). For every integer  $m \geq 1$ , we have

$$H^0(\mathbb{P}_{\mathbb{Z}}^n, \mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(m)) = \bigoplus_{|\alpha|=m} \mathbb{Z}X^\alpha.$$

Let  $t = \sum_{|\alpha|=m} a_\alpha X^\alpha \in H^0(\mathbb{P}_{\mathbb{Z}}^n, \mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(m))$  be a global section. Denote by  $M$  (resp.  $L$ ) the geometric vector bundle associated to  $\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(m)$  (resp.  $\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(1)$ ).

We first focus on the affine chart  $U_0 = \{X_0 \neq 0\}$  and let  $s = X_0 \in H^0(\mathbb{P}^n, \mathcal{O}(1))$ . For every point  $x = [1 : x_1 : \dots : x_n] \in U_0(\mathbb{C})$ ,

$$\|s(x)\| = \frac{1}{(1 + |x_1|^2 + \dots + |x_n|^2)^{1/2}} \implies \|t(x)\| = \frac{|\sum a_\alpha x^\alpha|}{(1 + |x_1|^2 + \dots + |x_n|^2)^{m/2}}.$$

Hence one can see that for any point  $x = [x_0 : \dots : x_n] \in \mathbb{P}^n(\mathbb{C})$ , we have

$$\|t(x)\| = \frac{|\sum a_\alpha x^\alpha|}{\|x\|^m}$$

with  $\|x\| = (|x_0|^2 + |x_1|^2 + \dots + |x_n|^2)^{1/2}$ .

In general, it seems difficult to compute  $h^0(\mathbb{P}_{\mathbb{Z}}^n, \overline{\mathcal{O}(m)})$  explicitly. For example, when  $m = 2$ , we have that  $\sup_{x \in \mathbb{P}^n(\mathbb{C})} \|t(x)\|$  is equal to the largest norm of eigenvalues of the matrix associated to the quadratic form  $(a_{ij})$ .

Yang: To be checked.

## Appendix